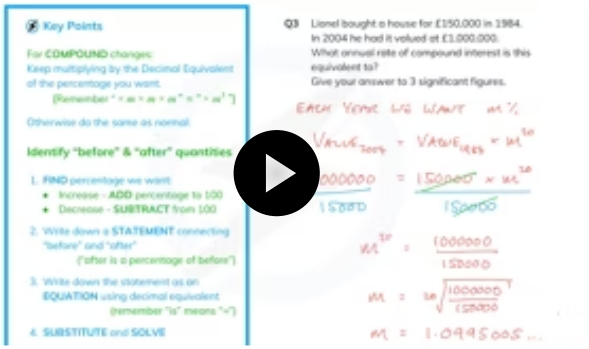
Compound Interest

Exam code: 0607

Download PDF

Written by: Jamie Wood
Reviewed by: Dan Finlay

Updated on 20 January 2026



Did this video help you? Yes No

Compound Interest

What is compound interest?

- **Compound interest** is where interest is calculated on the **running total**, not just the starting amount
  - This is different from **simple interest** where interest is only based on the starting amount
- E.g. **$ 100** earns **10% interest** each year, for 3 years
  - At the end of year 1, **10% of $ 100 is earned**
    - The total balance will now be 100+10 = **$ 110**
  - At the end of year 2, **10% of $ 110 is earned**
    - The balance will now be 110+11 = **$ 121**
  - At the end of year 3, **10% of $ 121 is earned**
    - The balance will now be 121+12.1 = **$ 133.10**

How do I calculate compound interest?

- Compound interest **increases an amount by a percentage** and then **increases the new amount by the same percentage**
  - This process repeats each time period (yearly or monthly etc)
- We can use a **multiplier** to carry out the percentage increase multiple times

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

would find (300×1.05)×1.05
        - This could be rewritten as **300×1.05^2**
    - To increase **$ 300** by **5%**, each year for **3 years**, we would find ((300×1.05)×1.05)×1.05
        - This could be rewritten as **300×1.05^3**
- This can be extended to any number of periods that the interest is applied for
    - If **$ 2000** is subject to **4%** compound interest each year for **12 years**
    - Find **2000×1.04^{12}**, which is $ 3202.06
- Note that this method calculates the **total balance** at the end of the period, **not the interest earned**

Compound interest formula

- An **alternative method** is to use the **following formula** to calculate the final balance
    - Final balance = $P\left(1+\frac{r}{100}\right)^n$ where
        - *P* is the original amount,
        - *r* is the % increase,
        - and *n* is the number of years
    - Note that $1+\frac{r}{100}$ is the same value as the multiplier
        - e.g. 1.15 for 15% interest
- This formula is **not given** in the exam

What if there are different rates of interest?

- Instead of applying the same percentage for n years, banks may offer **different rates**
    - E.g. 3% for 2 years and then 2% for 1 year
- Treat the periods under different rates as **separate calculations**
- For example if a sum of $500 earns **4% interest for 2 years**, and then **2.5% for 1 year**
    - For the first period
        - **500×1.04^2** = 540.80
    - For the second period
        - **540.80×1.025** = 554.32
    - So the total amount is now **$554.32** at the end of the 3 years

How do I solve reverse compound interest problems?

- You could be **told the final balance after** compound interest has been applied, and **need to find the original amount**
    - This could be referred to as a "**reverse compound interest**" problem
- For example if:
    - The final balance is £432
    - After 20% interest has been applied each year

- $432 = P \times 1.20^3$ where P is the original amount
- Solve for P,
  - Divide both sides by 1.20^3
  - $432 \div 1.20^3 = P$
  - $P = £250$

- In general, to find the original amount:
  - Divide the final amount by m^n where
    - m is the multiplier for the time period
    - and n is the number of time periods (usually years)

Examiner Tips and Tricks

- Double check if the question uses **simple** interest or **compound** interest
- The formula for compound interest is **not** given in the exam

Worked Example

Jasmina invests $1200 in a savings account, which pays compound interest at the rate of 4% per year for 7 years.

To the nearest dollar, what is her investment worth at the end of the 7 years?

Answer:

Method 1

We want an increase of 4% per year
This is equivalent to a multiplier of 1.04, or 104% of the original amount

This multiplier is applied 7 times
$\times 1.04 \times 1.04 \times 1.04 \times 1.04 \times 1.04 \times 1.04 \times 1.04 = 1.04^7$

Therefore the final value after 7 years will be

$$1200 \times 1.04^7 = \$1579.118135...$$

Round to the nearest dollar

$$\$1579$$

Method 2

Using the formula for the final amount $P\left(1 + \frac{r}{100}\right)^n$
Substitute P is 1200, r = 4 and n = 7 into the formula

$$1200\left(1 + \frac{4}{100}\right)^7$$

$$\$1579$$

Worked Example

Martin invested $\$n$ in an account for 3 years at a rate of 3.9%.

At the end of the 3 years the balance of the account is $3925.67.

Find the value of n to the nearest dollar.

Answer:

n is increased by 3.9% for 3 years, resulting in 3925

Write an equation describing this

$$n \times 1.039^3 = 3925.67$$

Divide both sides by 1.039^3

$$n = \frac{3925.67}{1.039^3}$$

Work this out on your calculator

$$n = 3499.992764...$$

Round to the nearest dollar

$$n = 3500$$

Teacher only

Test your students on this in just a few clicks

- Create exam-aligned tests
- Tailor format & difficulty
- Quizzes or homework in minutes

Try Test Builder →

Test yourself

Was this revision note helpful? Yes No

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.
Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.